PREPVECTOR
Cognitive Assessment Platform

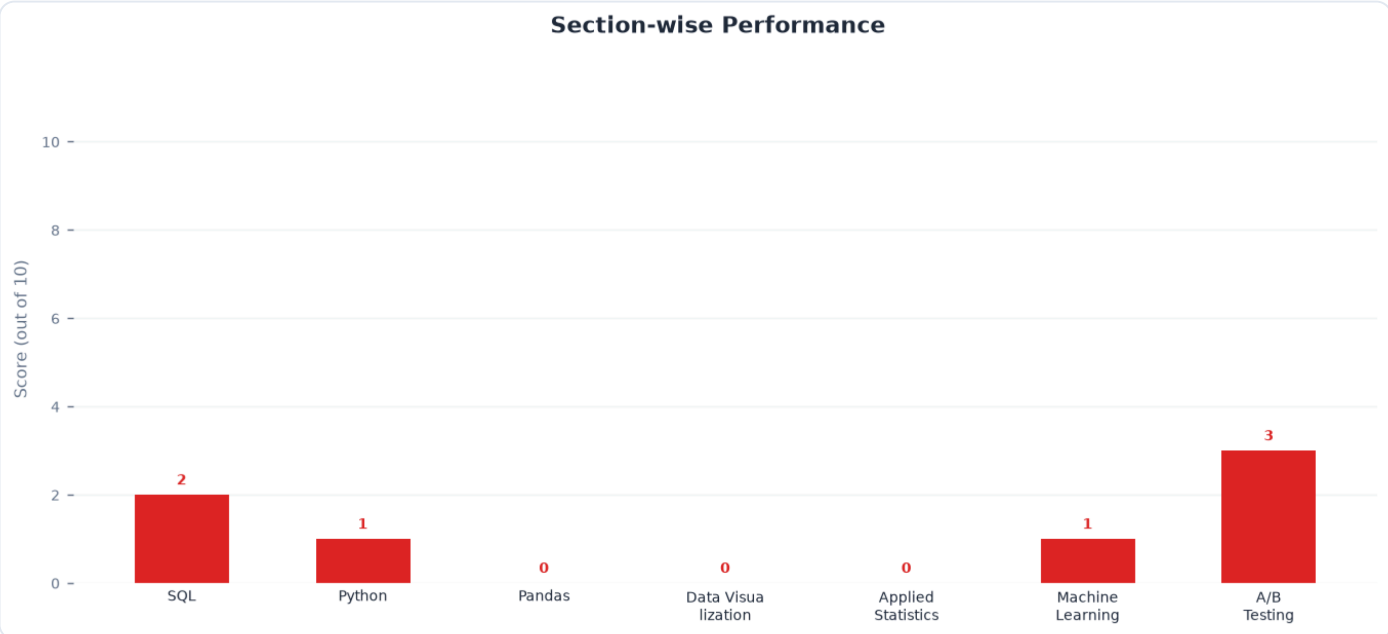
AI Evaluation Report

Data Science Expert — Self-Assessment

CANDIDATE	FINAL SCORE	CORRECT ANSWERS
Julie Anderson	**11** / 70 weighted points	**7** / 35 questions NEEDS IMPROVEMENT

Assessment Date: July 25, 2026 **Report Generated:** July 25, 2026 at 20:41 UTC **Role:** Data Science Expert

OVERALL RATING	SUCCESS RATE	WEIGHTED SCORE	TOP SKILL	NEEDS IMMEDIATE ATTENTION
★☆☆☆☆	**15.7%**	**11 / 70**	**A/B Testing**	**Pandas**



Skill Rating & Benchmark

Section	Your Score	Reference Benchmark (Demo)	Delta	Status
SQL	2 / 10	6.8	-4.8	Critical Gap
Python	1 / 10	6.2	-5.2	Critical Gap
Pandas	0 / 10	5.8	-5.8	Critical Gap
Data Visualization	0 / 10	6.0	-6.0	Critical Gap
Applied Statistics	0 / 10	5.5	-5.5	Critical Gap
Machine Learning	1 / 10	5.2	-4.2	Critical Gap
A/B Testing	3 / 10	6.4	-3.4	Needs Work

Attempt Statistics Breakdown

Section	Total Q	Attempted	Correct	Incorrect	Skipped	Accuracy
SQL	5	5	2	3	0	40%
Python	5	5	1	4	0	20%
Pandas	5	5	0	5	0	0%
Data Visualization	5	5	0	5	0	0%
Applied Statistics	5	5	0	5	0	0%
Machine Learning	5	5	1	4	0	20%
A/B Testing	5	5	3	2	0	60%

Executive Summary

The candidate answered 7 out of 35 questions correctly, resulting in a weighted score of 11 out of 70. This performance indicates critical gaps across core data science competencies, particularly in advanced SQL, statistical diagnostics, and Python execution flow. A structured, hands-on upskilling plan is highly recommended to meet industry standards.

Key Strengths

- **Basic Python & Pandas Familiarity:** Shows awareness of core concepts like decorators, closures, and DataFrame manipulation, though execution details require refinement.
- **A/B Testing Concepts:** Recognizes fundamental experimental design terms, indicating exposure to product analytics frameworks.

- **Machine Learning Evaluation:** Understands the structural components of confusion matrices and SHAP values, providing a foundation for explainable AI.

Top Blind Spots

- **Advanced SQL & Query Performance:** Gaps in understanding window function mechanics (specifically `ROW_NUMBER` tie-breaking) and how composite indexes optimize queries via index seeks versus scans.
- **Statistical Diagnostics & OLS Assumptions:** Inability to correctly diagnose model violations from residual plots (confusing heteroscedasticity with non-linearity) and Q-Q plots (misidentifying heavy-tailed distributions).
- **Python Scope & Execution Flow:** Misunderstanding how closures bind variables to maintain state in memory and the exact sequential execution of decorated functions.

Actionable Learning Resources

SQL Performance: Complete the "Advanced SQL" and "Query Tuning" pathways on Mode Analytics, focusing on window functions and index-seek optimization.

✓ Action Item

Statistical Diagnostics: Read Chapter 3 (Linear Regression) of *An Introduction to Statistical Learning (ISLR)*, specifically focusing on residual plots, heteroscedasticity, and Q-Q plot interpretation.

✓ Action Item

Python Closures & Decorators: Work through the "Python Closures" and "Primer on Python Decorators" deep-dive tutorials on Real Python to master scope and state preservation.

✓ Action Item

A/B Testing & Network Effects: Study the "Udacity A/B Testing" course by Google, focusing on SUTVA violations, network interference, and sample size calculations.

✓ Action Item